

## ADVANCED ANALYTICS, MACHINE LEARNING, DEEP LEARNING & AI COMPREHENSIVE INTERVIEW QUESTION BANK

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### SECTION 1: ARTIFICIAL INTELLIGENCE FUNDAMENTALS

1. Define Artificial Intelligence.
  2. Differentiate between AI, Machine Learning, and Deep Learning.
  3. Explain Narrow AI, General AI, and Super AI.
  4. Types of Machine Learning: Supervised, Unsupervised, Reinforcement Learning.
  5. Explain Bias-Variance tradeoff.
  6. Define Overfitting and Underfitting.
  7. What is Data Leakage?
  8. Explain Cross-Validation techniques.
  9. What is Feature Engineering?
  10. Explain Model Generalization.
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### SECTION 2: MATHEMATICS & STATISTICS FOR ML

Probability & Statistics 1. Central Limit Theorem. 2. Hypothesis Testing and p-value. 3. Type I and Type II Errors. 4. Covariance vs Correlation. 5. Bayes Theorem with example. 6. Likelihood vs Probability. 7. Normal Distribution and its properties. 8. Multicollinearity.

Linear Algebra 1. Vectors and Matrices in ML. 2. Eigenvalues and Eigenvectors. 3. Role of Eigenvalues in PCA. 4. Matrix Rank and Determinant. 5. Singular Value Decomposition. 6. Dot Product and its significance.

Calculus 1. Role of derivatives in optimization. 2. Gradient Descent algorithm. 3. Types of Gradient Descent. 4. Learning Rate impact. 5. Convex vs Non-Convex functions.

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### SECTION 3: SUPERVISED LEARNING

Linear Regression 1. Assumptions of Linear Regression. 2. Cost function derivation. 3. Multicollinearity problem. 4. Ridge vs Lasso Regression. 5. Regularization techniques.

Logistic Regression 1. Sigmoid Function. 2. Log Loss. 3. Decision Boundary. 4. Why MSE is not suitable for classification.

Decision Trees & Ensembles 1. Entropy and Information Gain. 2. Gini Index. 3. Overfitting in Trees. 4. Pruning techniques. 5. Random Forest algorithm. 6. Gradient Boosting. 7. XGBoost concept.

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### SECTION 4: UNSUPERVISED LEARNING

1. K-Means Algorithm and steps.
2. Elbow Method.
3. Silhouette Score.

4. Hierarchical Clustering.
  5. DBSCAN.
  6. Principal Component Analysis.
  7. Importance of Scaling before PCA.
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## SECTION 5: DEEP LEARNING

Neural Networks 1. Perceptron model. 2. Activation Functions (ReLU, Sigmoid, Tanh). 3. Forward and Backpropagation. 4. Vanishing and Exploding Gradient problems. 5. Loss Functions.

CNN 1. Convolution operation. 2. Padding and Stride. 3. Pooling layers. 4. Transfer Learning.

RNN and LSTM 1. Limitations of RNN. 2. LSTM architecture. 3. GRU. 4. Attention Mechanism. 5. Transformer architecture basics.

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## SECTION 6: MODEL EVALUATION

1. Confusion Matrix.
  2. Accuracy, Precision, Recall.
  3. F1 Score.
  4. ROC and AUC.
  5. RMSE and MAE.
  6. R-Squared.
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## SECTION 7: SCENARIO-BASED QUESTIONS

1. Handling Imbalanced Data.
  2. Model performs well in training but poorly in testing.
  3. Designing a Customer Churn Prediction system.
  4. Fraud Detection Model Design.
  5. End-to-end ML pipeline design.
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## SECTION 8: CODING ROUND

1. Implement Linear Regression using sklearn.
  2. Perform Data Preprocessing and Feature Scaling.
  3. Hyperparameter Tuning using GridSearch.
  4. Build a CNN using PyTorch or TensorFlow.
  5. Implement Gradient Descent from scratch.
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## SECTION 9: MLOPS & DEPLOYMENT

1. What is MLOps?
2. Model Drift and Data Drift.

3. CI/CD in ML.
4. Model Monitoring strategies.
5. Docker for ML deployment.

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